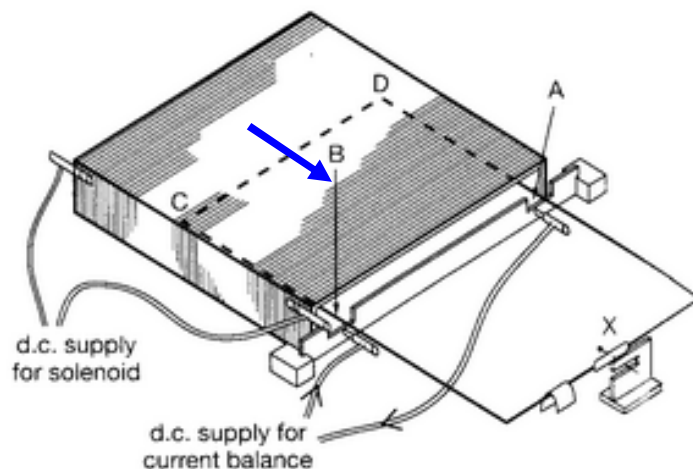


- 4 (a) (i) V-x graph:
 Potential constant within both spheres [1]
 Maximum potential for -Q (twice) higher than that of +Q [1]
 Correct shape (hyperbola) and cut at 6 cm. [1]
- (ii) E-x graph:
 Graph is negative [1]
 Magnitude of E at -Q is higher than that at +Q. [1]
 Correct shape (hyperbola) [1]
 As the spheres are taken to be point charges, the turning point of the graph is at x = 6 cm, which is in the middle of the centres of both spheres. [1]

- (b) Volume, V, of each drop $\frac{4}{3}\pi r^3$, hence, $V \propto r^3$
 After combining, the volume of new drop is 2V, hence,
 $V \propto r^3 \Rightarrow \left(\frac{r'}{r}\right)^3 = \frac{2V}{V} \Rightarrow r' = (\sqrt[3]{2})r$ [1]

Potential, $V = \frac{Q}{4\pi\epsilon_0 r}$, hence,
 $V \propto \frac{Q}{r} \Rightarrow \frac{V'}{V} = \left(\frac{Q'}{Q}\right)\left(\frac{r}{r'}\right) = \left(\frac{2Q}{Q}\right)\left(\frac{r}{(\sqrt[3]{2})r}\right)$ [1]
 $V' = \frac{2}{\sqrt[3]{2}}(500) = 793.7 \text{ V} = 790 \text{ (3 sf)}$ [1]

(c) (i)



(ii) $F = BIL = (0.035)(6.0)(0.15) = 0.0315 \text{ N}$ [1]

(iii) The perpendicular distance between the knife edge and each end is d .
Taking moments about **AB**,

$$Mg = 0.00315$$

$$M = 0.00321 \text{ kg}$$
 [1]

(iv) Align the section **CD** so that it is parallel to the external magnetic field. [1]
There will be no moment caused by this external magnetic field. [1]

